

Book Metadata API Documentation

1. Overview

The Book Metadata API is a RESTful API that allows the user to browse and query book metadata. It provides full CRUD operations for books, authors, and categories as well as analytical endpoints for searching, filtering and ranking books by rating and popularity.

- **Base URL:** <https://book-metadata-api.onrender.com>
 - **Response Format:** All responses are returned as JSON.
-

2. Authentication

- **GET requests** are public and require no authentication.
- **POST, PATCH, and DELETE requests** require an **Authorization** header.

Header

```
Authorization:  
8d9a1fe6425324282d852de957b6d09dfa1373928022351034d2560f00af476f
```

Example

```
curl -X 'POST' \  
  'https://book-metadata-api.onrender.com/books/' \  
  -H 'accept: application/json' \  
  -H 'Authorization:  
8d9a1fe6425324282d852de957b6d09dfa1373928022351034d2560f00af476f' \  
  -H 'Content-Type: application/json' \  
  -d '{ ... }'
```

3. Endpoints

Books

POST /books/

Create a new book record.

Authentication: Required

Request Body

Field	Type	Required	Description
title	string	Yes	Title of the book
author_id	integer	No	ID of the author
category_id	integer	No	ID of the category
description	string	No	Short description of the book
rating	float	No	Rating between 0.0 and 5.0
views	integer	No	Number of views (default: 0)

Example Request

```
curl -X 'POST' \
  'https://book-metadata-api.onrender.com/books/' \
  -H 'accept: application/json' \
  -H 'Authorization:
8d9a1fe6425324282d852de957b6d09dfa1373928022351034d2560f00af476f' \
  -H 'Content-Type: application/json' \
  -d '{
  "title": "string",
  "average_rating": 0,
  "ratings_count": 0,
  "author_ids": [],
  "category_ids": []
}'
```

Example Response 201 Created

```
{
  "title": "string",
  "average_rating": 0,
  "ratings_count": 0,
  "id": 1001,
  "authors": [],
  "categories": []
}
```

GET /books/

Retrieve a list of all books.

Authentication: Not required

Query Parameters

Parameter	Type	Required	Default	Description	Max
skip	integer	No	0	Number of records to skip	None
limit	integer	No	100	Maximum number of records to return	500

Example Request

```
curl -X 'GET' \
  'https://book-metadata-api.onrender.com/books/?skip=0&limit=2' \
  -H 'accept: application/json'
```

Example Response 200 OK

```
[
  {
    "title": "Harry Potter and the Half-Blood Prince (Harry Potter #6)",
    "average_rating": 4.57,
    "ratings_count": 2095690,
    "id": 1,
    "authors": [
      {
        "id": 1,
        "name": "J.K. Rowling"
      },
      {
        "id": 2,
        "name": "Mary GrandPré"
      }
    ],
    "categories": []
  },
  {
    "title": "Harry Potter and the Order of the Phoenix (Harry Potter #5)",
    "average_rating": 4.49,
    "ratings_count": 2153167,
    "id": 2,
    "authors": [
      {
        "id": 1,
        "name": "J.K. Rowling"
      },
      {
        "id": 2,
        "name": "Mary GrandPré"
      }
    ],
    "categories": []
  }
]
```

```
    }  
  ]  
}
```

GET /books/{book_id}

Retrieve a single book by its ID.

Authentication: Not required

Path Parameters

Parameter	Type	Description
book_id	integer	ID of the book

Example Request

```
curl -X 'GET' \  
  'https://book-metadata-api.onrender.com/books/1' \  
  -H 'accept: application/json'
```

Example Response 200 OK

```
{  
  "title": "Harry Potter and the Half-Blood Prince (Harry Potter #6)",  
  "average_rating": 4.57,  
  "ratings_count": 2095690,  
  "id": 1,  
  "authors": [  
    {  
      "id": 1,  
      "name": "J.K. Rowling"  
    },  
    {  
      "id": 2,  
      "name": "Mary GrandPré"  
    }  
  ],  
  "categories": []  
}
```

PATCH /books/{book_id}

Update one or more fields of an existing book.

Authentication: Required

Path Parameters

Parameter	Type	Description
book_id	integer	ID of the book

Request Body *(all fields optional)*

Field	Type	Description
title	string	Updated title
average_rating	float	Updated average rating
ratings_count	integer	Updated ratings count
author_ids	array of integers	Replaces the full list of associated authors
category_ids	array of integers	Replaces the full list of associated categories

Example Request

```
curl -X 'PATCH' \
  'https://book-metadata-api.onrender.com/books/2' \
  -H 'accept: application/json' \
  -H 'Authorization:
8d9a1fe6425324282d852de957b6d09dfa1373928022351034d2560f00af476f' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Harry"
  }'
```

Example Response 200 OK

```
{
  "title": "Harry",
  "average_rating": 4.49,
  "ratings_count": 2153167,
  "id": 2,
  "authors": [
    {
      "id": 1,
      "name": "J.K. Rowling"
    },
    {
      "id": 2,
      "name": "Mary GrandPré"
    }
  ],
  "categories": []
}
```

DELETE /books/{book_id}

Delete a book by its ID.

Authentication: Required

Path Parameters

Parameter	Type	Description
book_id	integer	ID of the book

Example Request

```
curl -X 'DELETE' \
  'https://book-metadata-api.onrender.com/books/5' \
  -H 'accept: */*' \
  -H 'Authorization: 8d9a1fe6425324282d852de957b6d09dfa1373928022351034d2560f00af476f'
```

Example Response 204 No content

(empty body)

GET /books/top-rated/

Retrieve books sorted by highest average rating. Only books with a non-null **average_rating** are included. Use **min_ratings** to filter out books with too few ratings to be meaningful.

Authentication: Not required

Query Parameters

Parameter	Type	Required	Default	Description	Max
skip	integer	No	0	Number of records to skip	None
limit	integer	No	100	Maximum number of records to return	500
min_ratings	integer	No	0	Minimum ratings_count required to be included	None

Example Request

```
curl -X 'GET' \
  'https://book-metadata-api.onrender.com/books/top-rated/?
skip=0&limit=2&min_ratings=1000' \
  -H 'accept: application/json'
```

Example Response 200 OK

```
[
  {
    "title": "Harry Potter Collection (Harry Potter #1-6)",
    "average_rating": 4.73,
    "ratings_count": 28242,
    "id": 7,
    "authors": [
      {
        "id": 1,
        "name": "J.K. Rowling"
      }
    ],
    "categories": []
  },
  {
    "title": "J.R.R. Tolkien 4-Book Boxed Set: The Hobbit and The Lord of the
Rings",
    "average_rating": 4.59,
    "ratings_count": 101233,
    "id": 22,
    "authors": [
      {
        "id": 7,
        "name": "J.R.R. Tolkien"
      }
    ],
    "categories": []
  }
]
```

GET /books/most-popular/

Retrieve books sorted by highest `ratings_count`. Only books with a non-null `ratings_count` are included.

Authentication: Not required

Query Parameters

Parameter	Type	Required	Default	Description	Max
skip	integer	No	0	Number of records to skip	None

Parameter	Type	Required	Default	Description	Max
limit	integer	No	100	Maximum number of records to return	500

Example Request

```
curl -X 'GET' \
  'https://book-metadata-api.onrender.com/books/most-popular/?skip=0&limit=2' \
  -H 'accept: application/json'
```

Example Response 200 OK

```
[
  {
    "title": "Angels & Demons (Robert Langdon #1)",
    "average_rating": 3.89,
    "ratings_count": 2418736,
    "id": 308,
    "authors": [
      {
        "id": 302,
        "name": "Dan Brown"
      }
    ],
    "categories": []
  },
  {
    "title": "Harry Potter and the Prisoner of Azkaban (Harry Potter #3)",
    "average_rating": 4.56,
    "ratings_count": 2339585,
    "id": 4,
    "authors": [
      {
        "id": 1,
        "name": "J.K. Rowling"
      },
      {
        "id": 2,
        "name": "Mary GrandPré"
      }
    ],
    "categories": []
  }
]
```

GET /books/search/

Search books by title keyword (case-insensitive partial match).

Authentication: Not required

Query Parameters

Parameter	Type	Required	Default	Description	Max
keyword	string	Yes	_	Keyword to search in book titles	—
limit	integer	No	100	Maximum number of records to return	500
min_ratings	integer	No	0	Minimum ratings_count required to be included	None

Example Request

```
curl -X 'GET' \
  'https://book-metadata-api.onrender.com/books/search/?
keyword=mockingbird&skip=0&limit=2' \
  -H 'accept: application/json'
```

Example Response 200 OK

```
[
  {
    "title": "To Kill a Mockingbird",
    "average_rating": 4.27,
    "ratings_count": 10524,
    "id": 799,
    "authors": [
      {
        "id": 784,
        "name": "Harper Lee"
      }
    ],
    "categories": []
  },
  {
    "title": "To Kill a Mockingbird",
    "average_rating": 4.27,
    "ratings_count": 697,
    "id": 800,
    "authors": [
      {
        "id": 784,
        "name": "Harper Lee"
      },
      {
        "id": 785,
        "name": "Sissy Spacek"
      }
    ],
  },
]
```

```
    "categories": []
  }
]
```

GET /books/filter/author/{author_id}

Retrieve all books associated with a specific author.

Authentication: Not required

Path Parameters

Parameter	Type	Description
author_id	integer	ID of the author

Query Parameters

Parameter	Type	Required	Default	Description	Max
skip	integer	No	0	Number of records to skip	None
limit	integer	No	100	Maximum number of records to return	500

Example Request

```
curl -X 'GET' \
  'https://book-metadata-api.onrender.com/books/filter/author/67?skip=0&limit=2' \
  -H 'accept: application/json'
```

Example Response 200 OK

```
[
  {
    "title": "The Power of One: One Person  One Rule  One Month",
    "average_rating": 4.28,
    "ratings_count": 16,
    "id": 76,
    "authors": [
      {
        "id": 65,
        "name": "John C. Maxwell"
      },
      {
        "id": 66,
        "name": "Stephen R. Graves"
      },
      {

```

```
      "id": 67,
      "name": "Thomas G. Addington"
    }
  ],
  "categories": []
}
```

GET /books/filter/category/{category_id}

Retrieve all books associated with a specific category.

Authentication: Not required

Path Parameters

Parameter	Type	Description
category_id	integer	ID of the category

Query Parameters

Parameter	Type	Required	Default	Description	Max
skip	integer	No	0	Number of records to skip	None
limit	integer	No	100	Maximum number of records to return	500

Example Request

```
curl -X 'GET' \
  'https://book-metadata-api.onrender.com/books/filter/category/1?skip=0&limit=2' \
  -H 'accept: application/json'
```

Example Response 200 OK

```
[
  {
    "title": "Harry Potter and the Half-Blood Prince (Harry Potter #6)",
    "average_rating": 4.57,
    "ratings_count": 2095690,
    "id": 1,
    "authors": [
      {
        "id": 1,
        "name": "J.K. Rowling"
      },
    ],
  },
]
```

```
    {
      "id": 2,
      "name": "Mary GrandPré"
    },
    {
      "id": 1,
      "name": "Fiction"
    }
  ],
  "categories": [
    {
      "id": 1,
      "name": "Fiction"
    }
  ]
}
```

Authors

POST /authors/

Create a new author record.

Authentication: Required

Request Body

Field	Type	Required	Description
name	string	Yes	Full name of the author

Example Request

```
curl -X 'POST' \
  'https://book-metadata-api.onrender.com/authors/' \
  -H 'accept: application/json' \
  -H 'Authorization: 8d9a1fe6425324282d852de957b6d09dfa1373928022351034d2560f00af476f' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Iven"
  }'
```

Example Response 201 Created

```
{
  "id": 972,
  "name": "Iven"
}
```

GET /authors/

Retrieve a list of all authors.

Authentication: Not required

Query Parameters

Parameter	Type	Required	Default	Description	Max
skip	integer	No	0	Number of records to skip	None
limit	integer	No	100	Maximum number of records to return	500

Example Request

```
curl -X 'GET' \
  'https://book-metadata-api.onrender.com/authors/?skip=0&limit=2' \
  -H 'accept: application/json'
```

Example Response 200 OK

```
[
  {
    "id": 1,
    "name": "J.K. Rowling"
  },
  {
    "id": 2,
    "name": "Mary GrandPré"
  }
]
```

GET /authors/{author_id}

Retrieve a single author by their ID.

Authentication: Not required

Path Parameters

Parameter	Type	Description
author_id	integer	ID of the author

Example Request

```
curl -X 'GET' \
  'https://book-metadata-api.onren
```

Example Response 200 OK

```
{
  "id": 4,
  "name": "Douglas Adams"
}
```

PATCH /authors/{author_id}

Update an existing author record.

Authentication: Required

Path Parameters

Parameter	Type	Description
author_id	integer	ID of the author

Request Body *(all fields optional)*

Field	Type	Description
name	string	Updated name

Example Request

```
curl -X 'PATCH' \
  'https://book-metadata-api.onrender.com/authors/5' \
  -H 'accept: application/json' \
  -H 'Authorization:
8d9a1fe6425324282d852de957b6d09dfa1373928022351034d2560f00af476f' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": "Neumann Otter"
}'
```

Example Response 200 OK

```
{
  "id": 5,
```

```
  "name": "Neumann Otter"
}
```

DELETE /authors/{author_id}

Delete an author by their ID.

Authentication: Required

Path Parameters

Parameter	Type	Description
author_id	integer	ID of the author

Example Request

```
curl -X 'DELETE' \
  'https://book-metadata-api.onrender.com/authors/666' \
  -H 'accept: */*' \
  -H 'Authorization:
8d9a1fe6425324282d852de957b6d09dfa1373928022351034d2560f00af476f'
```

Example Response 204 No Content

(empty body)

Categories

POST /categories/

Create a new category.

Authentication: Required

Request Body

Field	Type	Required	Description
name	string	Yes	Name of the category

Example Request

```
curl -X 'POST' \
  'https://book-metadata-api.onrender.com/categories/' \
  -H 'accept: application/json' \
  -H 'Authorization:
8d9a1fe6425324282d852de957b6d09dfa1373928022351034d2560f00af476f' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Fantasy"
  }'
```

Example Response

```
{
  "id": 2,
  "name": "Fantasy"
}
```

GET /categories/

Retrieve a list of all categories.

Authentication: Not required

Query Parameters

Parameter	Type	Required	Default	Description	Max
skip	integer	No	0	Number of records to skip	None
limit	integer	No	100	Maximum number of records to return	500

Example Request

```
curl -X 'GET' \
  'https://book-metadata-api.onrender.com/categories/?skip=0&limit=2' \
  -H 'accept: application/json'
```

Example Response

```
[
  {
    "id": 1,
    "name": "Fiction"
  },
  {
```

```
    "id": 2,
    "name": "Fantasy"
  }
]
```

GET /categories/{category_id}

Retrieve a single category by its ID.

Authentication: Not required

Path Parameters

Parameter	Type	Description
category_id	integer	ID of the category

Example Request

```
curl -X 'GET' \
  'https://book-metadata-api.onrender.com/categories/1' \
  -H 'accept: application/json'
```

Example Response 200 OK

```
{
  "id": 1,
  "name": "Fiction"
}
```

PATCH /categories/{category_id}

Update an existing category.

Authentication: Required

Path Parameters

Parameter	Type	Description
category_id	integer	ID of the category

Request Body *(all fields optional)*

Field	Type	Description
-------	------	-------------

Field	Type	Description
name	string	Updated category name

Example Request

```
curl -X 'PATCH' \
  'https://book-metadata-api.onrender.com/categories/1' \
  -H 'accept: application/json' \
  -H 'Authorization:
8d9a1fe6425324282d852de957b6d09dfa1373928022351034d2560f00af476f' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Fantasy"
  }'
```

Example Response 200 OK

```
{
  "id": 1,
  "name": "Fantasy"
}
```

DELETE /categories/{category_id}

Delete a category by its ID.

Authentication: Required

Path Parameters

Parameter	Type	Description
category_id	integer	ID of the category

Example Request

```
curl -X 'DELETE' \
  'https://book-metadata-api.onrender.com/categories/1' \
  -H 'accept: */*' \
  -H 'Authorization:
8d9a1fe6425324282d852de957b6d09dfa1373928022351034d2560f00af476f'
```

Example Response 204 No Content

(empty body)

4. Error Codes

Status Code	Name	Description
200	OK	Request succeeded. Returned for successful GET and PATCH requests.
201	Created	Resource successfully created. Returned for successful POST requests.
204	No Content	Resource successfully deleted. No body is returned.
401	Unauthorized	Missing or invalid Authorization header on a protected endpoint.
404	Not Found	The requested resource does not exist.
422	Unprocessable Entity	Request body failed validation (e.g. missing required field, wrong type).

Error Response Format

All errors return a JSON body with a **detail** field:

For **401** authentication errors:

```
{
  "detail": "Invalid or missing API key"
}
```

For **404** resource error:

```
{
  "detail": "Book not found"
}
```

For **422** validation errors:

```
{
  "detail": [
    {
      "type": "missing",
      "loc": [
        "body",
```

```
    "title"  
  ],  
  "msg": "Field required",  
  "input": {  
    "average_rating": 0,  
    "ratings_count": 0,  
    "author_ids": [],  
    "category_ids": []  
  }  
}  
]  
}
```